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Countermeasures for cardiac fibrosis in space travel: It takes more than a towel for a hitchhiker's guide to the galaxy

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Abstract

MicroRNAs (miRNAs) are small, non-coding RNA molecules that are post-transcriptional regulators of gene expression. miRNAs have been shown to be key regulators of complicated pathological processes and hence great biomarkers for the early prediction of diseases, such as cardiovascular diseases and radiation-associated alteration after spaceflight. In this study, we present possible antagomir treatments targeting three different miRNAs, miR-16-5p, miR-125b-5p, and let-7a-5p, to mitigate the activity of the spaceflight environment in cardiovascular diseases. We focus on three proteins of interest associated with fibrotic remodeling, TGF-β1, SMAD3, and COL1, analyzing the molecular outcomes of antagomir treatment when exposed to Galactic Cosmic Radiation (GCR), Solar Particle Events (SPE) radiation, and microgravity. These proteins have been shown to play different fibrotic and antifibrotic roles and show molecular changes associated with exposure to the space environment. Furthermore, our results demonstrate the therapeutic potential of antagomirs as a countermeasure for future spaceflight missions.

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Introduction

The strive for manned space missions has regained much of its former popularity and political attention in the last years. While multiple space agencies have entered the competition for the first manned Mars mission, the world is also about to enter an era of civilian and commercial spaceflight [1, 2, 3]. Along with an expected rise of manned space missions, special emphasis lays on the medical requirements for space travel to guarantee safe and successful missions [4]. In the absence of gravity, multiple adaptation processes of the human body have been reported. Along with an expected rise of manned space missions, special emphasis lays on the medical requirements for space travel to guarantee safe and successful missions [5]. While virtually all organs are affected by the environmental changes associated with space travel, in particular the cardiovascular system deserves closer attention in this context, since it is responsible for the majority of potential medical complications encountered in space travel [6, 7]. Short-term environmental challenges during space travel arise predominantly from changing gravity (hypo-, partial and hypergravity) and the psychological stressors: confinement and isolation. However, along with hemodynamic changes in response to weightlessness, special emphasis lies on cosmic radiation, especially concerning deep space missions. The impact of these effects on the cardiovascular system has been the topic of a plethora of studies since the start of the first space missions [8, 9, 10, 11]. More recently, miRNAs have gained a larger focus in spaceflight experiments.

Micro ribonucleic acids (miRNAs) are short non-coding RNA sequences with approximately 18–24 nucleotides in length that are single-stranded and are characterized by their distinctive “hairpin-loop” structure. They have been found in most eukaryotic organisms, acting in post-transcriptional regulation of gene expression and protein translation [12, 13]. They are essential for maintenance of physiological conditions, such as differentiation, proliferation and apoptosis of various cell types within the body [14, 15]. However, miRNAs have also been identified to play crucial roles in the regulation and prediction of various diseases, such as cancer [16], neurodegenerative diseases [17, 18], autoimmune disorders [19], and cardiovascular diseases [20, 21]. In addition to that, previous studies have unveiled the influence of microgravity [22, 23] and ionizing radiation (IR) [23, 24] on miRNA expression. In modeled microgravity it has been shown that 42 miRNAs were dysregulated, most notably miR-9-3p, miR-155-5p, miR-150-3p, and miR-378-3p that are also associated with genes involved in the inflammatory response, apoptosis and cell proliferation[22]. Additionally, the combination of microgravity and IR was shown to alter let-7i*, miR-7, miR-7-1*, miR-27a, miR-144, miR-200a, miR-598, miR-650 levels, suggesting a crucial role of miRNAs in cellular defense against oxidative and exogenous stress [23, 24]. In order to unveil the complex actions of miRNAs in response to spaceflight while simultaneously reducing the risk for study participants, it is crucial to identify the miRNAs that humans and rodents have in common. Beheshti et al. have previously identified 13 miRNAs that are associated with a systemic response due to spaceflight by computational screening of spaceflown rodents [25]. Additionally, transforming growth factor beta (TGF-β1) was hypothesized to be an overall regulator of miRNA expression, as all miRNAs identified were directly influenced by TGF-β1 levels. In turn, TGF-β1 was shown to be significantly mediated by microgravity and space radiation [25, 26]. As TGF-β1 plays fundamental roles in several different pathologic remodeling conditions, such as carcinogenesis [27], tissue fibrosis [28, 29, 30], and heart diseases [31, 32], it may also be a key protein in the systemic response to microgravity [25]. More recently, deep space radiation was performed on human peripheral blood mononuclear cells (PBMCs) serum samples. Furthermore, serum samples of astronauts were screened for these miRNAs by single-cell RNA sequencing. In their study, the sharing of a defined miRNA signature in humans and rodents was proven [33]. During the last decades miRNAs and their roles have become more important as diagnostic and potential therapeutic targets. MiRNA antagonists, the so-called antimirs or antagomirs, have been developed to specifically silence distinct miRNAs and their function in order to protect from oncologic diseases [34, 35], infections [36], as well as cardiovascular diseases [20, 37]. Most recently, Malkani et al. [33] highlighted the role of antagomirs in the damage caused by simulated deep space radiation. They successfully restored the human 3D vascular constructs of deep space irradiated samples by targeting miR-125, miR-16, and let-7a by appropriate antagomirs [33].

Considering the need for effective preventive measures especially in the context of long duration deep space missions such as journeys to Mars, a deeper understanding of the impact of cosmic radiation on the cardiovascular system and miRNA expression is of utmost importance. Cardiac fibrosis is driven by inflammatory cytokines, such as interleukins [38, 39], and growth factors, including the TGF-β superfamily [40, 41], such as the TGF-β isoforms (TGF-β1, β2, and β3), growth differentiation factors (GDFs), inhibins,

activins, and bone morphogenetic proteins (BMPs). In particular, as already reported for cardiac diseases, the TGF-β1 is the major driver of human pathologies [42]. As a response to an increased TGF-β1 signaling, myofibroblasts deposit excessive extracellular matrix (ECM) proteins [43, 44, 45]. Downstream, TGF-β1 initiates the phosphorylation and activation of suppressor of mothers against decapentaplegic (SMAD) transcription factors, such as SMAD2 and SMAD3 that induce the expression of profibrotic genes and the deposition of collagens [46]. Furthermore, TGF-β1 acts as a chemoattractant for inflammatory cells that drive tissue remodeling and the fibrotic process [43, 44, 45]. Consequently, in the present study we analyzed the TGF-β1 levels, as well as its downstream proteins SMAD3 and COL1 in normal loaded and hindlimb unloaded mice. Furthermore, the impact of GCR or SPE irradiation was evaluated in combination with an antagomir treatment targeting TGF-β, respectively.

Results

2.1 Key changes with fibrotic markers in human astronaut data

To explore if fibrosis might be an issue for astronauts during spaceflight, we utilized astronaut data from the Japanese Space Agency (JAXA) Cell-Free Epigenome (CFE) Study mission [47] with astronauts in space for 120 days and the first civilian commercial 3-day space mission referred to as Inspiration4 (i4) [48]. For both missions, RNA-sequencing was performed on the blood samples. Specifically, for the JAXA mission RNA-seq was conducted on plasma cell-free RNAs (cfRNAs) from 6 astronauts that were on the International Space Station (ISS) for 120 days from pre-flight, during flight, and post flight time points. The i4 mission involved 4 astronauts (two males and two females) that orbited at low Earth orbit (LEO) at 590 km above the atmosphere. As discussed in Overbery et al. [48], since they orbited the Earth at a higher altitude, the i4 astronauts experienced the equivalent accumulated dose of radiation as astronauts would experience on the ISS for approximately 9 months.

To observe if fibrotic processes are being promoted during spaceflight in astronauts, we compiled a list of specific fibrosis genes and also the related fibrosis pathways (**Table S1**). The Fibrotic pathways were determined from multiple pathways in MSigDB related to the fibrotic genes. We mapped these genes and pathways to both the CFE Study RNA-Seq data (Fig. 1) and the i4 scRNA-seq data from PBMCs (Fig. 2). The overall patterns of the specific fibrotic associated genes and pathways indicate that fibrosis might be a long term issue for the CFE Study astronauts (Fig. 1). When observing the pattern of the specific fibrosis genes we notice that *PDGFα*, *TGF-β1*, and *SMAD3* are upregulated during spaceflight compared to the pre-flight, while the rest of the specific fibrotic genes are downregulated. When comparing post-flight to pre-flight, we observe that with increasing time all of the genes, except for *PDGFα*, *TGF-β1*, and *SMAD3*, are upregulated compared to pre-flight.

When analyzing the specific pathways for the CFE Study data an interesting pattern arises that indicates potential change of fibrosis occurring long term. Given our previous studies linking TGF-β expression

patterns to spaceflight risk and the similarities in TGF-β patterns seen in the JAXA and i4 astronauts, we performed a functional enrichment analysis of cfRNA TGF-β gene clusters (Fig. 2). Three primary gene clusters based on overall patterns from the hierarchical clustering were analyzed using numerous functional databases. As expected, the predominant pathway represented across all three clusters was TGF-β signaling. However, there were distinct functional differences between the higher expressed post-flight genes (UP cluster, red) and the lower expressed post-flight genes (DOWN cluster, blue). Notably, the UP cluster was highly enriched for SMAD signaling in multiple data sources, along with a strong transcription factor signaling response, indicated by the nucleoplasm and transcription regulator complex (GO: cellular compartment) and transcription factor binding site enrichment for numerous zinc finger proteins (TRANSFAC). In addition, enrichment for genes that encode SMAD-containing protein-protein complexes were enriched only in the UP cluster (CORUM).

The UP late cluster (yellow) was less enriched; however, a persistent transcription factor signal was present in this cluster and histone deacetylase complex, suggesting epigenetic changes and later time points post-flight may impact TGF-β biology in astronauts. In contrast, the DOWN cluster was enriched for genes involved in integrins, cell/focal adhesion, and related protein kinase signaling. Interestingly, one of our candidate miRNAs previously targeted by our group to promote vascular health, mIR-125b-5p), was linked to the gene expression pattern observed in the UP cluster, suggesting a role for this miRNA, and other miRNAs, to TGF-β-related signaling in response to spaceflight

Interestingly, the i4 astronauts also exhibit a similar trend. Specific fibrosis genes and the genes in the related fibrosis pathways were overlapped less than 10% with DEGs at R + 1 from PBMCs and subpopulations (CD4 T, CD8 T, other T, B, NK, CD14 Mono, CD16 Mono, DC, other) (Fig. 3A). *SMAD3* pathway was the most sensitive pathway and mainly downregulated and Fibronectin pathway was the most stable pathway in PBMCs (Fig. 3A). DEGs percentage were generally similar across PBMCs and subpopulations (Fig. 3A). Among specific fibrosis genes, *FN1*, *SMAD3*, and *FGF1* expression were up-regulated at R + 1 and keep this elevated expression to the long-term compared to pre-flights in PBMCs (Fig. 3B). In case of *TNC*, *VIM*, and *TGF-β1*, the expressions were not increased at R + 1 but increased at the long-term compared to pre-flights in PBMCs (Fig. 3B). Interestingly, *TGF-β1* was up-regulated at the long-term (R + 45 and R + 82) in all of the subpopulations, even if it was not increased at R + 1 compared to pre-flights (Fig. 3B). In the case of *SMAD3*, the expression was increased at R + 1 and kept increasing at the long-term except for monocytes (Fig. 3B). While the rest of the specific fibrotic gene, some of the genes were not expressed in all subpopulations or the expression change pattern was not consistent. Among fibrosis related pathways, Wnt signaling was significantly up-regulated in most populations at long-term (R + 45 and R + 82) (Fig. 3C). Furthermore, *TGF-β* signaling was significantly up-regulated in CD14 monocytes at long-term (Fig. 3C).

2.2 Impact of the space environment on heart weight with and without miRNA based countermeasures

A previous study revealed a shared miRNA signature between astronauts and rodents when exposed to spaceflight environment[29]. Similarly, based on the analysis of the human astronaut data (Figs. 1–3) that fibrosis might be a concern in long-term missions, we performed simulated spaceflight experiments in murine models to determine if space radiation-associated fibrosis can occur in the short term and be potentially targeted by a miRNA-based countermeasure. The simulated spaceflight experiments were done at the NASA Space Radiation Laboratory (NSRL) at Brookhaven National Laboratory (BNL) [49, 50]. C57BL/6 female mice were exposed to a simplified Galactic Cosmic Ray (GCR) simulated irradiation at 0.5 Gy [49] and a simulated Solar Particle Event (SPE) irradiation at 1.0 Gy (see methods for details). The GCR irradiation simulates the total accumulated dose astronauts will receive in a roundtrip mission to Mars approximately lasting 1.5 years in deep space. The SPE irradiation simulates the high acute dose of proton irradiation at different energies that an astronaut will receive from a potential solar flare from the sun [8, 51]. To simulate microgravity with and without irradiation we performed hindlimb unloading (HU) (also referred to hindlimb suspension) experiments [33, 52, 53] and compared to normally loaded (NL) mice. Lastly, based on our previous work discussed in the introduction[33, 54] we applied antagonists or antagomirs to three circulating miRNAs (i.e. miR-16-5p, miR-125b-5p, and let-7a-5p) related to the cardiovascular health risks associated with spaceflight as a potential countermeasure to mitigate the space radiation and microgravity impact to mitigate fibrosis in the heart. The endpoint for all data was at 24 hours after irradiation. Although we are aware that fibrosis can occur long after irradiation, we were studying the immediate impact of space irradiation and future studies will be done to reveal the long term impact the space environment has on fibrosis development.

To analyze if treatment itself may have a measurable effect on heart samples, we compared heart weights between the respective treatment groups (Fig. 4). Analyses were conducted for non-adjusted heart weights as well as for heart weights adjusted to body weights. For non-adjusted heart weights, a significant increase for SPE + HU controls was observed, while a decrease in GCR non-HU controls was evident. For adjusted heart weights both findings remained significant. Additionally, non-irradiated HU mice in the treatment group displayed a significant reduction in adjusted heart weights.

2.3 Impact of the space environment on cardiac fibrotic remodeling and how a miRNA based countermeasure can restore fibrotic markers back to baseline

To determine potential fibrotic remodeling in murine cardiac tissue exposed to 0.5 Gy galactic cosmic radiation (GCR) or 1.0 Gy solar particle event (SPE) irradiation, we performed histological analysis using Masson's Trichrome staining. Microscopic analyses showed no sign of an elevated fibrotic content in the GCR or SPE irradiated myocardium in comparison to the Sham group. In addition, neither NL mice nor HU mice developed visible fibrotic alterations in the tissue observed (Fig. 5).

Quantification of the relative fibrotic area (RFA) of Sham and GCR or SPE irradiated mice, respectively, showed no major differences in the fibrotic content of the different study groups. Solely there was a significant decrease of the RFA from 0.530 ± 0.600% in HU Sham to 0.165 ± 0.088% ($p = 0.0466$) in HU 0.5 Gy GCR irradiated mice (Fig. 6). However, as the median RFA of any study group did not extend 1% of the total tissue sections analyzed, the clinical significance of the difference in RFA is limited.

Given that gross histological examination did not reveal any major alteration caused by irradiation, we next determined the protein expression of fibrosis-associated proteins (Figs. 7–9). Therefore, western blot analysis was performed by normalizing the proteins of interest against an appropriate housekeeping protein. Furthermore, in order to compare the data of different blots, a reference sample was loaded onto each gel. Afterwards, the individual normalized values were related to the reference sample, which was defined as 100%, resulting in the relative protein value (RPV), given in percentage.

Based on our previous knowledge that miR-16-5p, miR-125b-5p, and let-7a-5p targets TGF-β1, we further explored the impact of the space environment with and without the countermeasure treatment (Fig. 7a). Interestingly, TGF-β1 was significantly reduced in 0.5 Gy GCR irradiated mice in comparison to the associated Sham group ($p = 0.0078$). Furthermore, antagomir treatment of GCR irradiated NL mice showed an enhancement of TGF-β1 protein levels ($p = 0.0138$) in comparison to the untreated NL GCR irradiated mice. Therefore, antagomir treatment was found to restore TGF-β1 levels in GCR irradiated mice (Fig. 7b). However, in HU mice, we found no changes in expression levels of TGF-β1, neither in untreated nor in antagomir-treated mice (Fig. 7b). Conversely, while in 1.0 Gy SPE irradiated mice (Fig. 7c), TGF-β1 concentrations were shown to be decreased in comparison to NL Sham ($p = 0.0106$), antagomir treated irradiated NL mice did not show any changes in comparison to the NL Sham untreated control group or NL untreated, though 1.0 Gy SPE irradiated mice. Similarly, SPE irradiation resulted in a significant decrease of TGF-β1 levels in the antagomir treated mice ($p = 0.0034$). In summary, antagomir treatment was able to rescue the TFG-β1 protein levels within the heart of NL 0.5 Gy GCR irradiated mice but failed to recover TGF-β1 in NL 1.0 Gy SPE irradiated mice.

As playing an important part in the TGF-β signaling cascade and the process of fibrotic remodeling, the concentration of SMAD3 was assessed (Fig. 8). As seen in Fig. 8b, antagomir treatment increased SMAD3 levels in Sham HU mice in comparison to Sham NL mice ($p = 0.0283$). On the other hand, GCR radiation in combination with the antagomir treatment significantly reduced SMAD3 protein levels in HU mice in comparison to the HU Sham Antagomir group ($p < 0.0001$). Also, antagomir treatment in HU mice reduced SMAD3 when compared to NL antagomir GCR irradiated mice ($p = 0.0052$). Similarly as in GCR irradiated mice, we observed an increase in SMAD3 protein levels in response to antagomir treatment (Fig. 8c). Furthermore, HU significantly elevated SMAD3 levels in comparison to the corresponding NL Sham group ($p = 0.0083$).

Given COL1 is the main protein characterizing fibrotic tissue, we target COL1 in our experimental groups of mice with or without antagomir treatment (Fig. 9). In the Sham NL group antagomir treatment seems to increase COL1 ($p < 0.0001$). Furthermore, GCR irradiation increased the COL1 content in NL mice in

comparison to Sham NL mice ($p < 0.0001$) while antagomir treatment seems to suppress COL1 protein levels in irradiated NL mice ($p < 0.0001$). GCR irradiation in untreated HU mice caused a decrease of COL1 in comparison to the NL GCR counterpart ($p < 0.0001$) with no effects of antagomir treatment. Interestingly, COL1 was significantly reduced in SPE irradiated mice in comparison to the Sham antagomir treated NL ($p < 0.0001$) and HU ($p < 0.0001$) mice. Notably, antagomir treatment had no effect on COL1 levels in both NL and HU groups.

Finally, we determined the association of the fibrosis-associated proteins with the area of fibrotic tissue measured by histological examination by correlating the irradiation and antagomir data with protein levels of TGF-β1, SMAD3, and COL1 within the heart. The results shown in Fig. 10 unveil that there is no direct correlation between the histological analyses with the levels of the fibrosis-associated proteins analyzed. Neither in GCR irradiated (Fig. 9a), nor in SPE irradiated mice (Fig. 9b) the protein contents were associated with fibrotic cardiac remodeling.

Discussion

Radiation exposure in space has always been a concern and risk factor for astronauts during and after spaceflight missions. Radiation can lead to damage in biological molecules, such as proteins and nucleic acids, either via energy absorption or Reactive Oxygen Species (ROS) generation as a result of the radiolysis of water molecules [55, 56]. Exposure to space radiation has been found to cause radiation-induced cardiovascular diseases leading to pathological consequences, for instance, myocardial remodeling and fibrosis [57, 58, 59, 60, 61].

Spaceflight-associated microRNA (miRNA) signatures have shown a promising contribution to regulating and predicting various cardiovascular system disruptions after spaceflight [59, 60]. Understanding the pathological impacts of space radiation and microgravity on cardiovascular systems and its associated miRNA-involved pathways is crucial for developing preventive or therapeutic approaches to improve spaceflight-associated myocardial fibrosis.

In previous studies, Beheshti et al. [25] identified 13 miRNAs that interacted with a master regulator (TGF-β1) of the microRNA response to space radiation and microgravity after space flight via unbiased systems biology analysis. However, the downstream pathway looking specifically at myocardial fibrosis had never been well studied together with a target treatment [25].

To determine the activity of the signaling pathways associated with the known miRNAs, we used blood samples collected from four crew members in the Inspiration 4 Mission and astronauts in the JAXA CFE Study and conducted single cell analysis, biochemical profiling, and a complete blood count. The fibrotic marker profile in the JAXA and Inspiration4 studies, demonstrated different fibrosis-associated genes and pathways affected by radiation in these spaceflight missions, being TGF-β1 the most influential to regulate miRNA expression.

Currently, there are no working countermeasures for cardiac alteration after spaceflight. Given our results indicating the miRNA's involvement in cardiac fibrotic remodeling, our team targeted three miRNAs associated with the TGF-β1 pathways, miR-16-5p, miR-125b-5p, and let-7a-5p. These specific, antagomir treatments should reveal their therapeutic potential in regulating miRNA expression, thereby improving tissue fibrosis in the cardiovascular system after spaceflight. We utilized hindlimb unloading and normal loading murine models, then irradiated with either 0.5 Gy GCR or 1.0 Gy SPE for 24 hours and treated these models with specific antagomirs. Although our data showed no noticeable fibrotic characteristics in heart tissues caused by radiation or microgravity when observed under the microscope (Fig. 5), the molecular quantification on fibrosis associated-proteins provides outstanding evidence of fibrotic remodeling in both microgravity and radiation conditions (Fig. 9b, c).

Following our analysis, we investigated fibrosis-associated proteins involved in miRNA regulation and expression, and their response to antagomir administration. Analysis of TGF-β1 expression levels suggested that antagomir treatment in NL GCR irradiated cardiac tissue can rescue the activity of TGF-β1 (Fig. 7a) bringing levels closely back to normal expression compared to the NL sham group. Contrarily, antagomir treatment of SPE irradiated cardiac tissue does not affect the relative expression of TGF-β1 in NL and HU groups compared to our sham control groups. Similarly, due to the analytical findings suggesting little to no significant changes in the HU GCR and SPE irradiated groups compared to our controls, we can suggest that antagomir treatment does not affect TGF-β1 levels of activity when microgravity is present in both GCR and SPE irradiated cardiac tissue; yet, it showed promising results as a treatment to rescue TGF-β1 activity in normal earth gravity conditions when GCR radiation is present.

The molecular quantification on COL1, a protein characterizing fibrotic tissue, demonstrated that 0.5 Gy GCR irradiated mice had elevated fibrotic characteristics compared to our control groups (Fig. 9b, 9c) demonstrating fibrotic remodeling in GCR radiation conditions. Correspondingly, being SMAD3 one of the major downstream regulators of the TGF-β1 pathway [62] involved in COL1 expression; we identified that these fibrosis-associated proteins drastically change either by increasing or decreasing their level of activity when exposed to GCR and SPE radiation in the space environment. Furthermore, we suggest that GCR affects cardiac tissue to a different extent than SPE radiation. In addition, based on our experimental results, microgravity effects are also independent of exposure to irradiation. While we demonstrate that antagomir treatment can rescue the activity of TGF-β1, it also decreased the relative expression of SMAD3. Given that miRNAs can regulate gene expression at a transcriptional and translation level targeting different genes [13], antagomir treatment could affect miRNAs regulating mRNA activity by inhibiting or promoting gene expression.

The limitations of our current study include the lack of observable histological fibrosis in mouse models and short exposure time under GCR or SPE radiation compared to longer spaceflight exposure time in other studies[22]. Our COL1 expression level analysis indicates the molecular change in fibrotic status in cardiac tissue after radiation, but no histological alteration was observed. Additionally, chronic radiation from space environment inducing cardiovascular diseases, such as myocardial remodeling and fibrosis, may appear as long-term effects developing more than 10 to 15 years after exposure [58, 59, 60]. Hence, the

observation window we set may be too short for fibrosis to develop to show pathological features in some irradiated mouse hearts. Although the loading of radiation administered in our mouse models was closely equivalent to mimic exposure received in deep space during a Mars mission, it is not representative enough of a real-life spaceflight mission with a different radiation dosage and a longer exposure time.

Therefore, for future directions, we suggest starting a trial in real spaceflight with a longer radiation window and exposure to a real microgravity environment, looking at the expression of other regulators of the TGF-β1 cascades, and analyzing antagomir activity, targeting miR-16-5p, miR-125b-5p, and let-7a-5p, to look into the specific effects on the different fibrosis-associated proteins. In the current study, our mouse model irradiated by GCR and SPE responded differently to the antagomir treatments regarding TGF-β1, SMAD3, and COL1 expression levels, indicating that GCR and SPE-associated pathways in fibrotic remodeling are likely different in cardiac tissue. Further study will be necessary to identify pathways triggered by different sources of radiation respectively and develop targeted antagomir treatments accordingly.

Methods

Inspiration4 astronaut sample collection

The exposure to radiation of the Inspiration4 mission crew was 2.2 mGy-Eq based on NASA Q quality factors and 50% FAX ModelEffective dose (Lens: 2.6 mGy, Skin: 2.7 mGy, Heart: 1.8 mGy, Central Nervous System: 1.3 mGy). Blood samples were collected before (L-92, L-44, and L-3 days) and after (R+1, R+45, and R+82 days) spaceflight. TGFβ pathway, Wnt pathway, FGF pathway, PDGFα pathway, RAAS pathway, SMAD3 pathway, Angiotensin II pathway, Fibronectin pathway, Specific fibrosis genes suggested in the paper has been used for identifying DEGs from I4 PBMCs and subpopulations. Seurat was used to normalize RNA count data and calculate average expression of each gene. The Seurat FindMarker function was used to calculate I4 DEGs ($P < 0.05$). Average expression of those genes was used to generate heatmap and heatmaps are generated by heatmap R package. For pathway enrichment analysis, fgsea package was used with I4 DEGs and selected pathways related to TGFb, Wnt, FGF, PDGFa, RAAS, SMAD3, Angiotensin II, and Fibronectin. All human studies were done with ethical approvals with established and approved IRBs at Weill Cornell Medicine.

JAXA Cell-Free Epigenome (CFE) Study transcriptomic data

Plasma CFE RNA differential expression data and study protocols were shared through NASA's GeneLab platform with accession number: OSD-530 [47]. Briefly, blood samples were collected from 6 astronauts before, during, and after the spaceflight on the ISS. Total RNA was purified from plasma samples and processed for RNA-seq analysis. Mean expression values were obtained from normalized read counts of 6 astronauts for each time point. Heatmaps were made using the custom made fibrotic genes and

pathways in **Table S1** on the normalized values per time point using R package pheatmap version 1.0.12. Functional cluster analysis on the TGFβ pathway was done by selecting individual gene clusters for functional analysis using the hierarchical clustering of the target gene set. Using the default analysis settings, we used the multi-query feature of g:Profiler [63] to identify functional enrichments within and across the clusters.

Murine simulated space environment experiments

15-week +/- 3-day old, C57Bl/6J wild-type (wt) female mice were purchased from Jackson Laboratories and housed at Brookhaven National Laboratory (BNL, Upton, NY). Upon arrival to BNL, mice were quarantined and acclimated to a standard 12:12 h light:dark cycle, with controlled temperature/humidity for 1-week prior to cage acclimation. Food and water were given ad libitum, and standard bedding was changed once per week. Mice were cage acclimated (n=10 mice per group; 2 mice per cage to maintain social interaction) 3-days prior to HU, followed by 14-days either normally loaded (NL) or hindlimb unloaded (HU, see details below). Irradiation was administered on day 13 and blood tissues were collected 24-hours post-irradiation and post-euthanasia by CO_2 overdose, followed by cervical dislocation. Blood was collected via the abdominal IVC in EDTA-coated tubes (0.5 M) and plasma was separated by centrifugation at 1,700Xg for 15 minutes. Plasma was collected and frozen at -80°C for storage. A 100 µl aliquot of cellular fraction was frozen and stored at -80°C for RNA analyses, while the remaining cellular fraction was lysed with 1x RBC lysis buffer (Thermo Fisher Scientific) followed by flow cytometric preparation and analyses, as described below. All organs were flash frozen and collected at the time of dissection. Specifically, the heart was cut in half vertically with the left half of the heart flash frozen and the right half the heart fixed in 4% paraformaldehyde. All tissues were stored at either -80°C or 4°C. Body weight tracking was performed on days -3, 0, 7 and 14.

Hindlimb unloading (HU) was performed using the adjusted Morey-Holton method for social housing (n=2 per cage) [52]. Briefly, mice were suspended from the tail using non-invasive traction tape attached to an adjustable pulley mounted on the top of a standard rat cage. The adjustable nature of the device allows the user to position the animal in a head-down position (approximately 30° to the horizon) once attached. The crossbar height, lateral crossbar position, and chain length are adjustable. The pulley is free to move along the crossbar that spans the length of the top of the cage enabling the mouse to navigate freely on one half of the cage using its forepaws and to interact with the second mouse in the cage, without enabling ambulation of the limbs. Individual food and water sources are provided to each mouse and replaced daily. Nestlets were provided to each mouse to encourage nesting. Control mice were housed in identical cages with normal ambulation.

On day 13, mice were transported from the Brookhaven Laboratory Animal Facility at BNL to the NASA Space Radiation Laboratory (NSRL) facility by animal care staff and were transferred to individual HU boxes. The following doses of irradiation were administered; simplified GCR sim (0.5 Gy), SPE (1 Gy), and Sham control (0 Gy). Mice were positioned in the plateau region of the Bragg curve and irradiated at

room temperature in individual HU boxes to enable whole body irradiation while maintaining hindlimb suspension. The NSRL physics staff performed the dosimetry. To simulate GCR, we used the simplified GCR simulation of ions, energy, and doses determined by a NASA consensus formula that consists of 5 ions: protons at 1000 MeV, ^{28}Si at 600 MeV/n, ^{4}He at 250 MeV/n, ^{16}O at 350 MeV/n, ^{56}Fe at 600 MeV/n, and protons at 250 MeV. This dose of radiation is equivalent to what an astronaut is predicted to receive in deep space during a Mars mission, though it is modeled as a single exposure over 25 minutes instead of the actual chronic exposures over 1.5 years. Further, GCR sim is a mixture of high and low LET ions in a ratio of 15% to 85%, respectively. To simulate SPE, we used a total dose of 1 Gy protons with energy ranges from 50 MeV to 150 MeV. For all irradiation treatments a 60x60 beam was utilized at NSRL. For radiation dose equivalence and biological reference, we used 5 Gy gamma irradiation with the cesium resource available at BNL, in the absence of HU due to resource limitations. Sham controls were treated similarly to GCR/SPE irradiated mice, including HU cage boxes and beam line (without irradiation) for the same duration as GCR simulation, i.e. 25 minutes.

Antagomir countermeasure treatment in the mice

Inhibitors, also referred to as antagomirs, to the following three specific miRNAs were purchased and designed from AUM biotech (Philadelphia, PA, USA): miR-16-5p, miR-125b-5p, and let-7a-5p. Specifically, the antagomirs were identified as AUM*antagomir* oligos. Each antagomir was given at a dose of 10 mg/kg combined for a total of 30 mg/kg total AUMantagomir dose by intraperitoneal (IP) injection every 3 days starting from 1 day prior to HU setup until 1 day prior irradiation.

Antibodies

The following antibodies were used in the analyses: Recombinant Anti-TGF beta 1 antibody [EPR18163] (1:1000 in dilution, abcam, Cambridge, UK), Anti-Smad3 antibody (1:1000 in dilution, abcam, Cambridge, UK), and Anti-Collagen I antibody (1:1000 in dilution, abcam, Cambridge, UK). Vinculin (E1E9V) XP® (1:2000 in dilution, Cell Signaling Technology, MA, USA), and GAPDH (D16H11) XP® (1:8000 in dilution, Cell Signaling Technology, MA, USA) were used as housekeeping proteins. Anti-rabbit IgG, HRP-linked (1:4000 in dilution, Cell Signaling Technology, MA, USA) served as a secondary antibody for immunoblot analyses.

Histological analysis

For histological analysis, formalin-fixed paraffin-embedded (FFPE) mouse cardiac tissue was cut into 6 µm thick serial sections and dried overnight. The next day, Masson's trichrome staining was performed to visualize the extent of fibrosis in cardiac tissue. In brief, tissue sections were deparaffinized by ROTI®Histol (Carl Roth, Germany) and rehydrated followed by the incubation with Bouin's fixative (Carl Roth, Germany) for ≥1 h at 60°C. Thereafter the following steps were performed to achieve the

appropriate staining: washing in dH_2O; incubating in Weigert's iron hematoxylin (Carl Roth, Germany); differentiation in 0.5% hydrogen chloride (HCl)/70% EtOH solution; bluing in tap water; incubation in Biebrich scarlet solution; washing in dH_2O; incubation in phosphomolybdic/phosphotungstic acid, followed by aniline blue; washing in dH_2O; incubating in 1% acetic acid, and a final washing step. Then the slices were dehydrated by an ethanol row finishing with 2 times Xylene (Carl Roth, Germany). Finally, the stained sections were mounted with Entellan® (Merck Millipore, MA, US) and dried overnight.

Microscopical analyses of the stained sections were performed using an inverse microscope Axiovert 200 (Axio Observer, Zeiss, Germany) and an EC Plan-NEOFLUAR 20x/0.5 objective (Zeiss, Germany). For quantitative analysis, depending on the sample size, 15 pictures were taken by Axiocam MRc3 (Zeiss, Germany) and AxioVision SE64 Rel. 4.8 (Zeiss, Germany) from each tissue sample to generate a mostly randomized collection of sections to be analyzed. To differentiate and quantify the fibrotic (blue) area in relation to the pure cardiac muscle areas (cardiomyocytes stained in red) per image, the software ImageJ (NIH, LOCI, US) was used. Therefore, a color threshold was set as follows: Hue 100-225, Saturation 0-255, and Brightness 0-180. Furthermore, to adequately assess the amount of red-colored myocardial tissue, a color deconvolution using Masson's Trichrome Vector was performed for each picture. The threshold was then set between 0-90 or 0-150, depending on the intensity of the staining. Within each sample stained, the same threshold was used. Finally, the relative fibrotic area (RFA) was calculated by dividing the pixels of the blue area by the pixels' total area of the tissue section, expressed as a percentage.

Immunoblotting

Shock frozen mouse cardiac specimen was lysed by RIPA lysis buffer system (150 mM NaCl, 50 mM Tris-HCl, 1% Nonidet P-40, 0.5% sodium deoxycholate, 0.1% SDS) with phenylmethylsulfonyl fluoride (PMSF) and protease inhibitor. Total protein concentration was determined using BCA assay (Thermo Fisher Scientific, NH, USA). Equal quantities of protein lysates were separated by SDS-PAGE on a 4-15% Mini-PROTEAN TGX Gel (Bio-Rad Laboratories, Austria) with 4x Laemmli buffer (Bio-Rad Laboratories, Austria) serving as the loading buffer. The separated proteins were further transferred to polyvinylidene fluoride (PVDF) Immun-Blot PVDF membranes (Bio-Rad Laboratories, Austria) by tank blotting. Membranes were blocked with 5% non-fat dry milk powder in TRIS-buffered saline with Tween-20 (TBS-T) for 1 h and were incubated overnight at 4°C, as indicated above. The next day, the membranes were incubated with a HRP-linked secondary antibody for 1 h at room temperature. The chemiluminescent signal was detected using a Raytest Stella 8005 (Elysia-Raytest, Germany) and the camera control software XStella Version 1.00.011 (Elysia-Raytest, Germany). In order to normalize the proteins of interest, the housekeeping proteins Vinculin or GAPDH were further incubated overnight at 4°C and the bands were detected the next day as indicated.

Data analysis and normalization were performed in Image Lab 6.0.1 software (Bio-Rad, Vienna, Austria) by generating a multichannel image. Thereafter, the lanes and bands were detected manually, and the

normalization channel (housekeeping proteins) and reference lane (reference sample) selected. Furthermore, the background subtraction was set to a disk size of 70.0 mm. The normalized quantities of the samples were calculated automatically and are given as the adjusted volume corrected by the normalization factor, named “Norm. Vol. (Int)”: Thereafter, the relative protein volume (RPV) for each sample was calculated in relation to the reference sample, which was termed as 100%. Consequently, the RPV of the proteins of interest may exceed 100% as the RVP is in relation to the 100% band of the reference sample.

Statistical analysis

The statistical analysis was performed using GraphPad PRISM 9 software (GraphPad-Software, San Diego, CA, USA). The data were tested for Gaussian approximation by Shapiro-Wilk test. Grouped analyses of the RFA and Western Blot results were performed using two-way ANOVA with Tukey’s multiple comparisons test. If the bar charts included normally distributed data, the figures are given as mean and standard deviation (SD); not normally distributed data is given as median and interquartile range (IQR; IQR=Q0.75−Q0.25). If the figures include both, normally distributed, as well as not normally distributed values, all datasets in the graphs are given as median and IQR. Correlation analyses were also performed according to the individual Gaussian approximation of the datasets. The nonparametric data was analyzed using Spearman correlation. In the figures of the correlation analyses, the simple linear regression lines are depicted. The alpha threshold and the confidence level were set as $p \leq 0.05$ for all analyses and the results are termed as statistically significant.

Declarations

Acknowledgements

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Author contributions

Conceptualization: A.B. and P.J.; Methodology: A.B., P.J., and V.P..; Formal Analysis: A.B., V.P., J.K., and J.C.S; Investigation: A.B., C.E.M., P.J., V.P., M.M., and J.S.C.; i4 data and omics: C.E.M., E.O., and J.K.; JAXA data: M.M., BNL Mouse Experiments: A.B., J.S.C., E.B., A.K., N.A., S.A., R.S.H., and L.E.O.; Resource: A.B., C.E.M., P.J., M.M., and J.C.S.; Original Draft: P.J., V.P., and A.B.; Review & Editing: all authors; Figures and Visualization: A.B., V.P., J.S.C., and J.K.; Funding Acquisition: A.B., C.E.M. (for i4 study), and P.J.; Supervision: A.B. and P.J.

Competing Interests

The Authors declare no Competing Financial or Non-Financial Interests

Data Availability

The JAXA CFE data is available via the NASA Open Science Data Repository's (OSDR)'s Biological Data Management Environment (https://osdr.nasa.gov/bio/repo) with accession numbers: OSD-530. The Inspiration 4 RNA-seq data will also be available at OSDR at the time of publication. All other data will be made available upon request.

The animal experiments performed for this manuscript were reviewed and approved by the Institutional Animal Care and Use Committee (IACUC) at Brookhaven National Laboratory under the protocol # 506.

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Figures

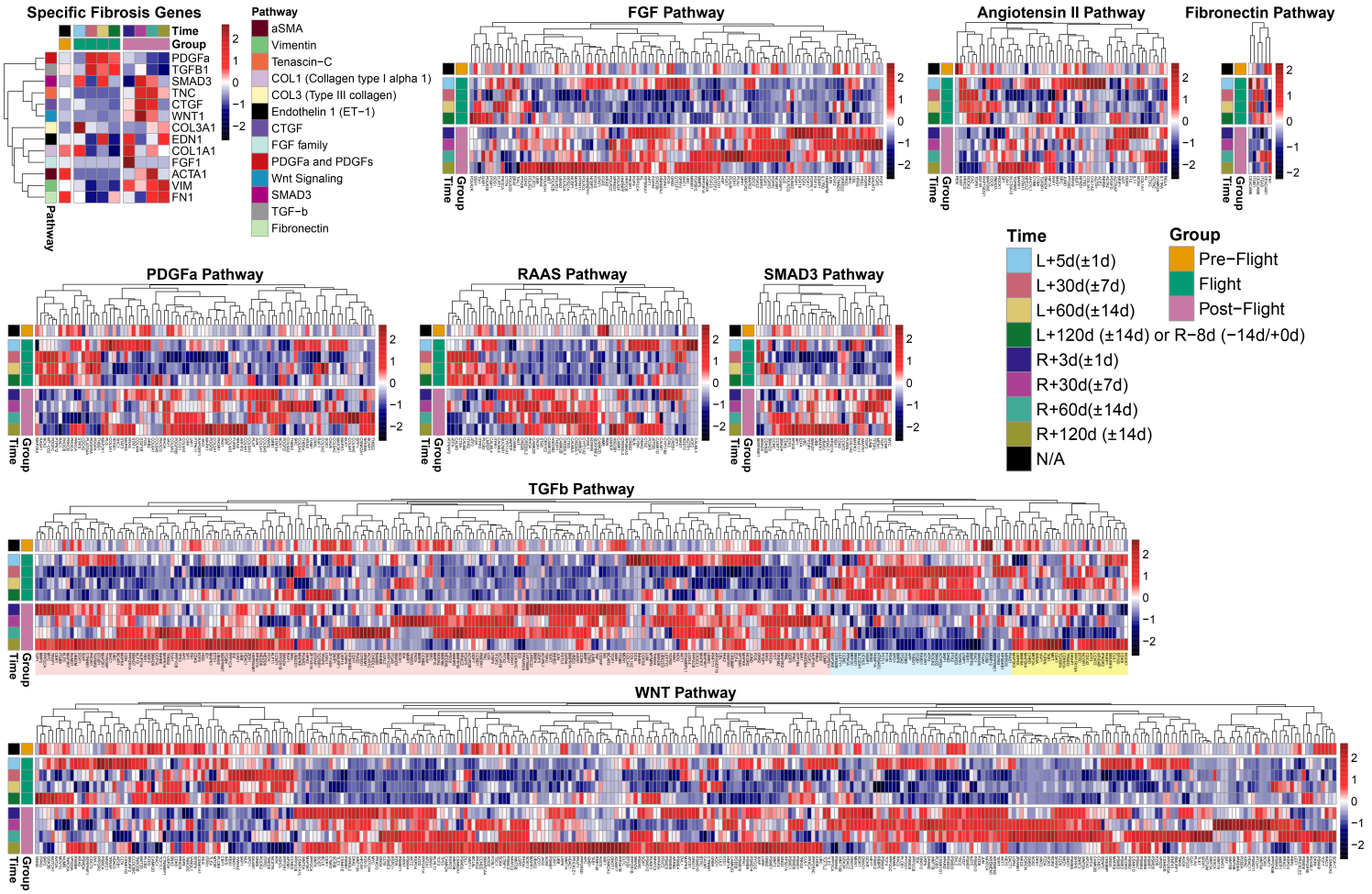


Figure 1

Fibrotic markers regulation in CFE Study astronaut plasma cfRNA. Heat map of specific Fibrotic genes and pathways being regulated in the six astronauts over 120 days in space. The values shown on are the averaged normalized expression values for all six astronauts each time point during flight and post-flight. The three pre-flight time points were averaged together, since the changes for genes in the time leading up to flight are considered to be the same and part of the baseline values. The specific genes for each fibrotic pathway are determined from various pathway databases and details can be found in **Table S1**. For the time, L = Launch (i.e. meaning time after launch from Earth and on the International Space

Station) and R = Return to Earth. The specific clusters of genes for the TGFβ pathway is color coded and further analysis is provided in **Fig. 2**.

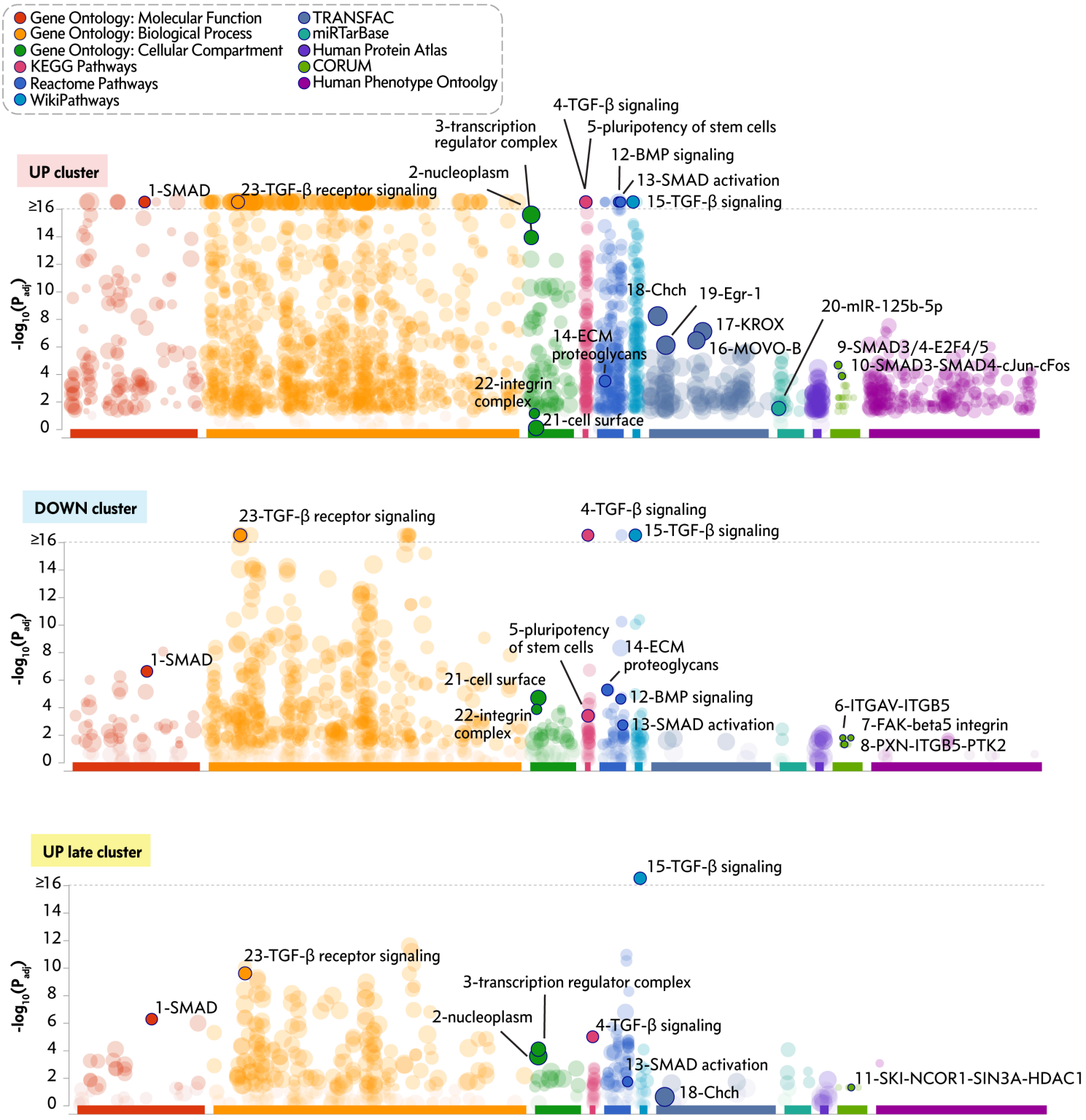


Figure 2

Functional changes in TGF-β-related genes. Enrichment analysis of genes from the three primary clusters of the TGF-β subsets (see **Fig. 1** TGFβ pathway) revealed the pathways, transcription factor binding sites, miRNAs, and protein-protein complexes common across or unique to each cluster.

Selected functional groups and their position across the three gene clusters are numbered and labeled. Data are represented by a dot plot using the -log10 of the adjusted P value (Benjamini-Hochberg false discovery rate) and grouped/colored by the indicated data source. The complete output is available in the data supplement (**Table S2**)

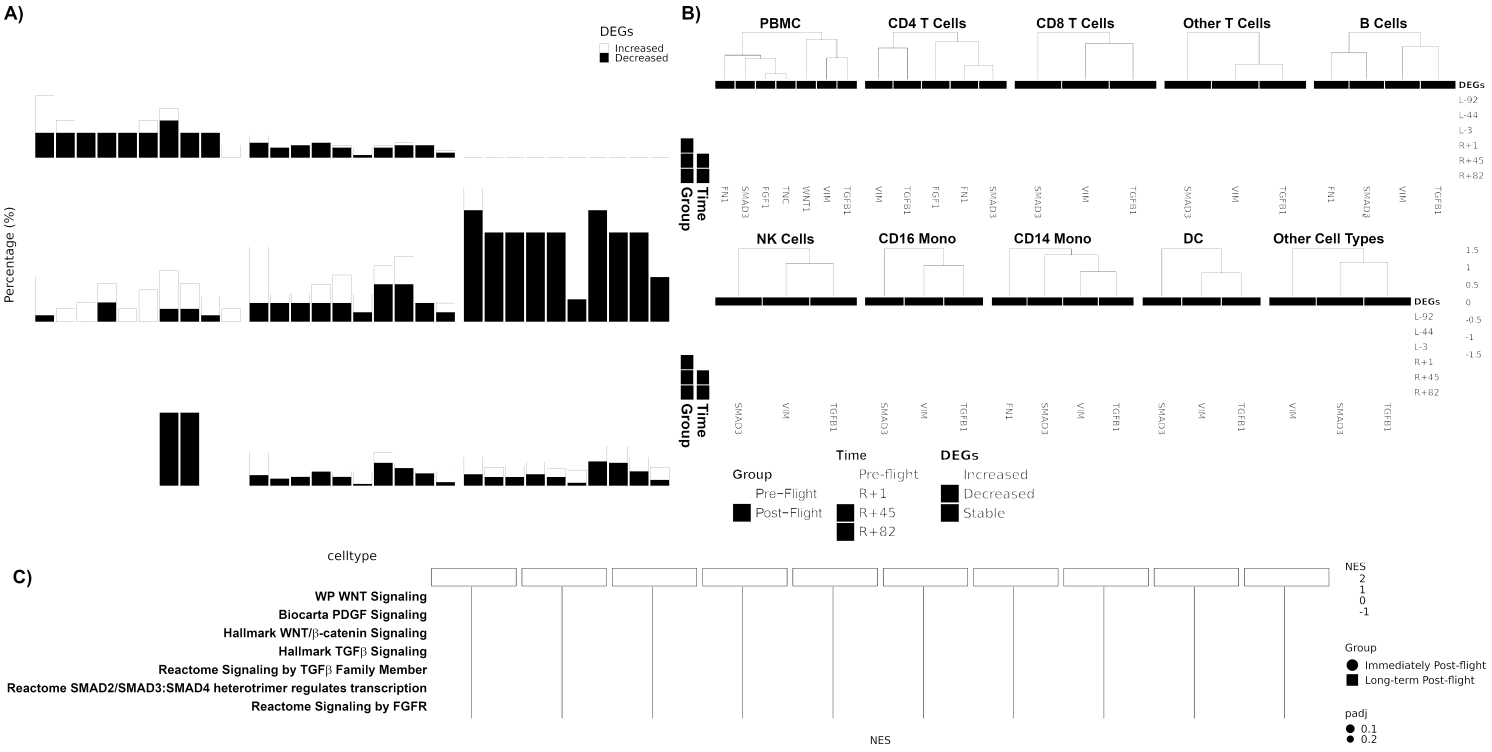
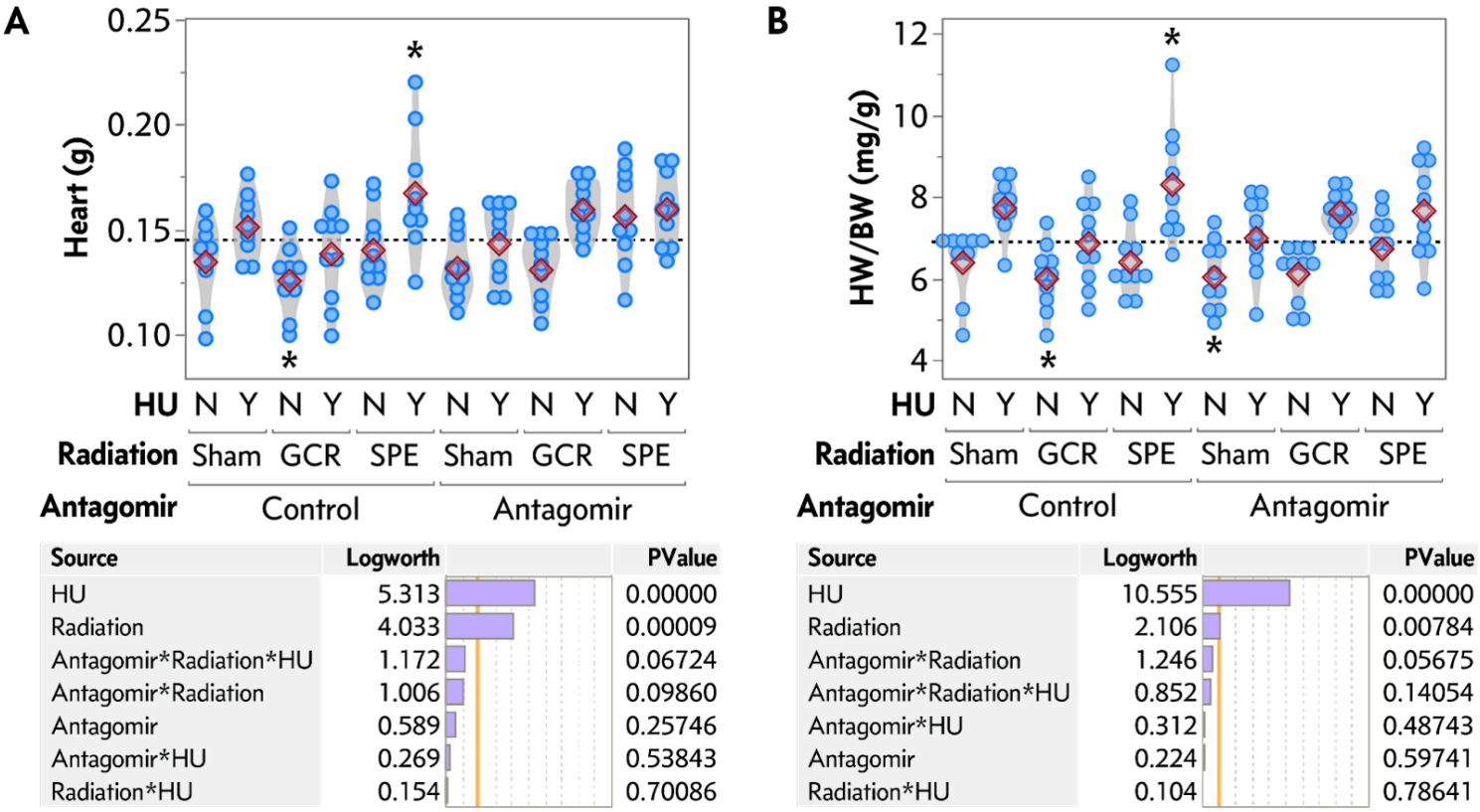


Figure 3

Fibrotic markers regulation in Inspiration4 (i4) astronaut blood cells. A) Bar plots showing the number of increase (or upregulated) and decreased (or downregulated) fibrotic genes for each fibrotic pathway in different cell types from single cell RNA (scRNA) sequence. **B)** Heatmaps of the expression values for the significantly regulated (p-value < 0.05) fibrotic genes in each cell type from the scRNA-seq data. **C)** Lollipop plots on the gene set enrichment analysis (GSEA) on the i4 scRNA-seq data with the custom fibrotic pathways. FDR < 0.3 was utilized as the cutoff for this analysis. The normalized enrichment score (NES) is plotted on the x-axis indicating if the pathway is predicted to be increased (red) or decreased (blue).



Source	Logworth	PValue
HU	5.313	0.00000
Radiation	4.033	0.00009
Antagomir*Radiation*HU	1.172	0.06724
Antagomir*Radiation	1.006	0.09860
Antagomir	0.589	0.25746
Antagomir*HU	0.269	0.53843
Radiation*HU	0.154	0.70086

Source	Logworth	PValue
HU	10.555	0.00000
Radiation	2.106	0.00784
Antagomir*Radiation	1.246	0.05675
Antagomir*Radiation*HU	0.852	0.14054
Antagomir*HU	0.312	0.48743
Antagomir	0.224	0.59741
Radiation*HU	0.104	0.78641

Figure 4

Fibrotic markers regulation in Inspiration4 astronaut blood cells. A) Raw heart weights. **B)** Adjust heart weights compared to the total body weight. The p-values are indicated in the tables below the plots. HU = hindlimb unloading, GCR = Galactic Cosmic Ray, and SPE = Solar Particle Event.

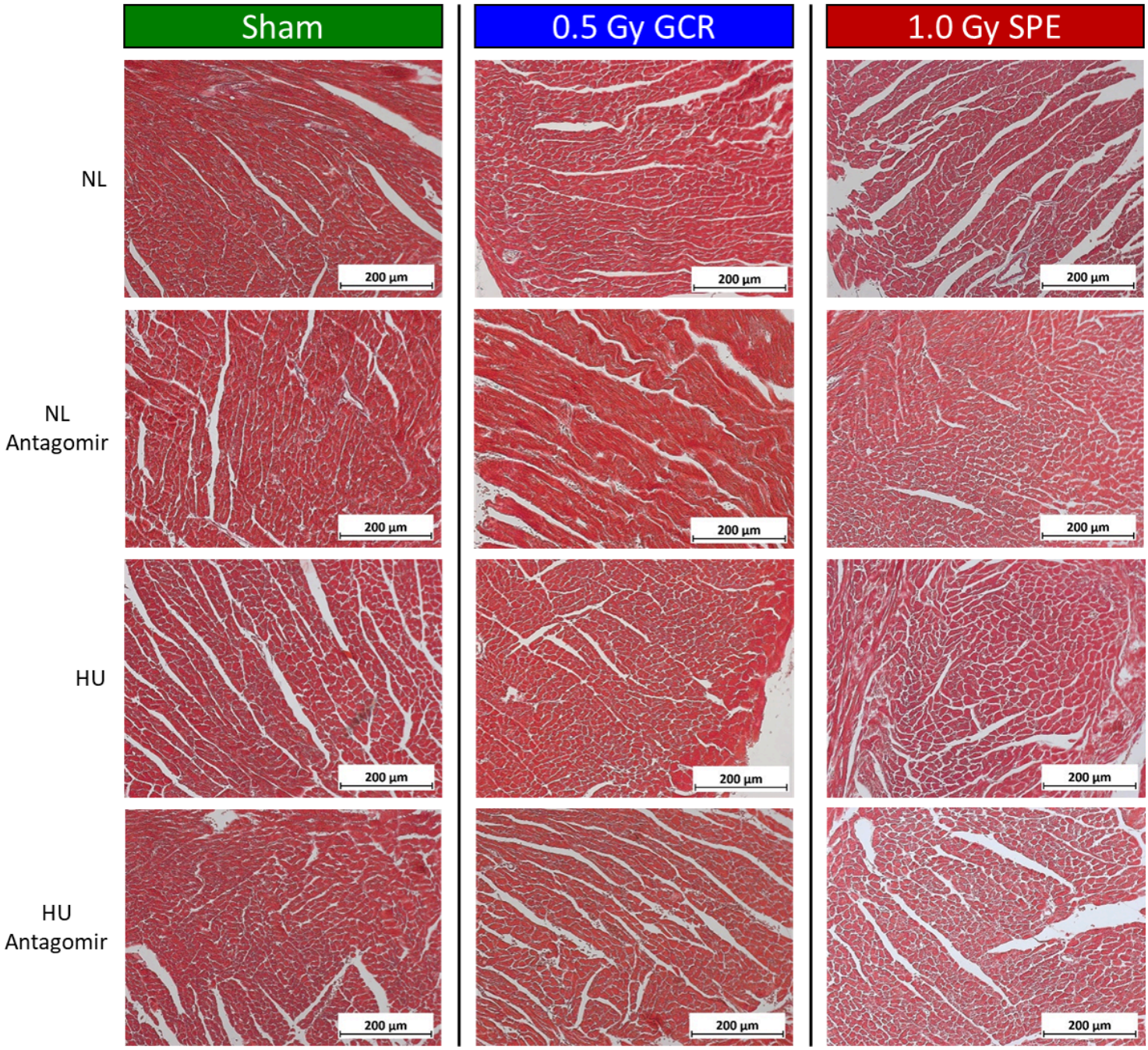


Figure 5

Histological assessment of the fibrotic content in GCR or SPE irradiated murine myocardium.
Cardiomyocytes are stained in red, fibrotic tissue is stained in blue. Magnification: 200x; GCR: galactic cosmic radiation; HU: hindlimb unloaded; NL: normal loaded; SPE: solar particle event

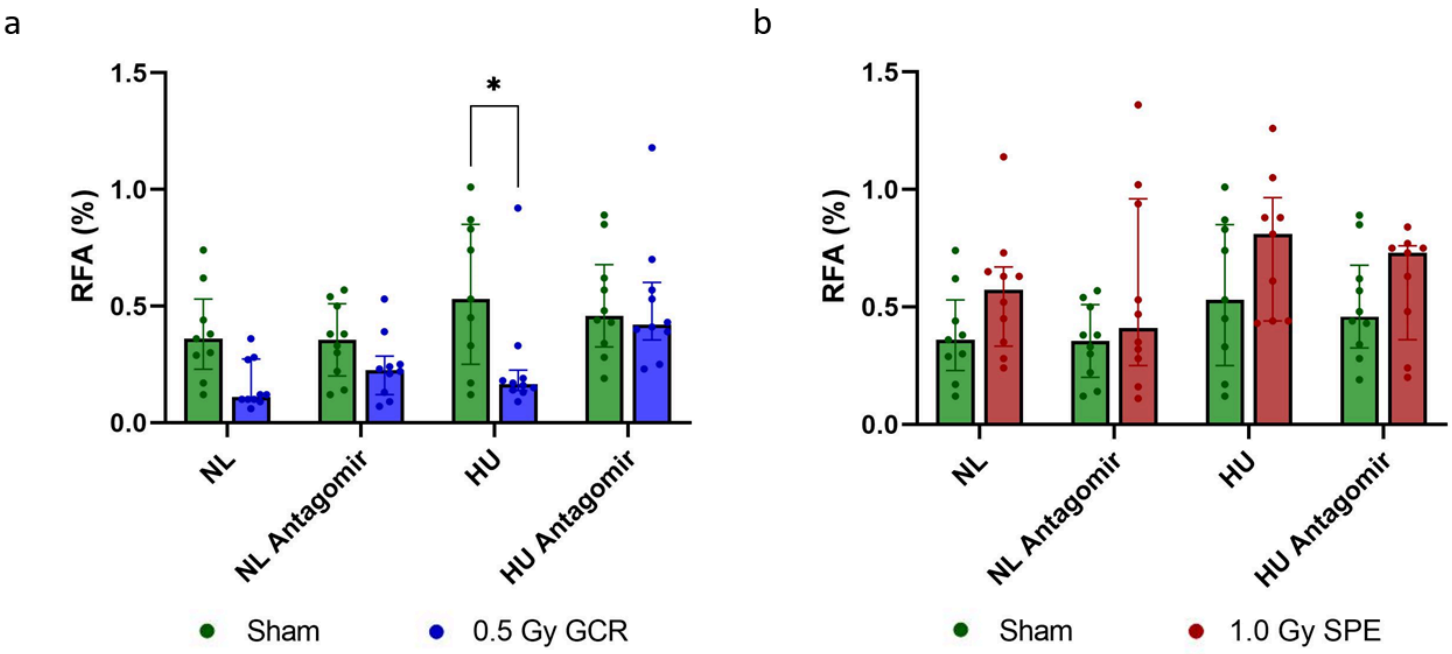


Figure 6

Quantification of the fibrotic content in (a) GCR irradiated and (b) SPE irradiated mice in comparison to Sham mice. The bar charts are given as median + IQR. GCR: galactic cosmic radiation; HU: hindlimb unloaded; IQR: interquartile range; NL: normal loaded; RFA: relative fibrotic area; SPE: solar particle event; * = $p<0.05$

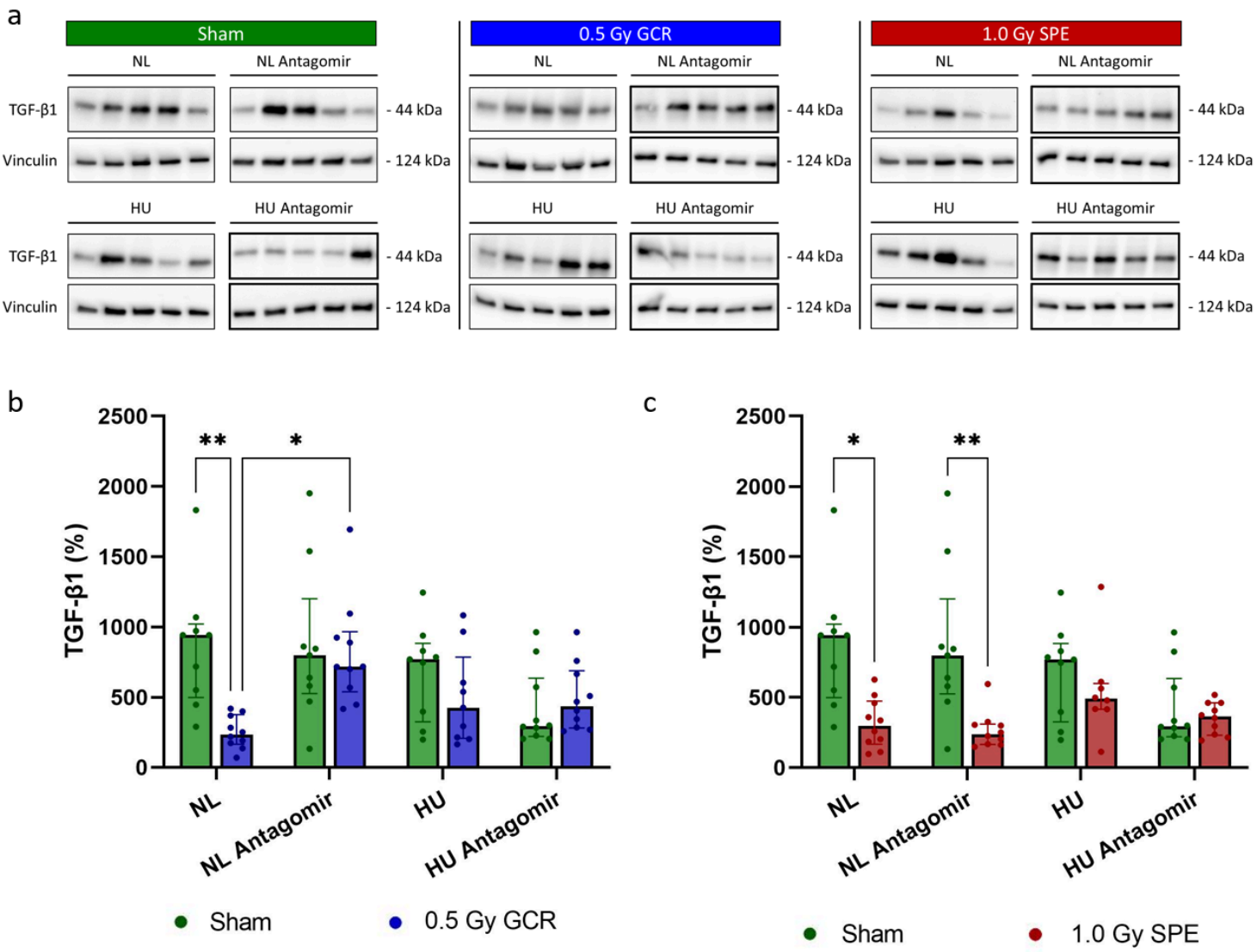


Figure 7

Quantification of TGF-β1 by western blot analysis. **(a)** Representative Western blot analysis of TGF-β1 in Sham and 0.5 Gy GCR or 1.0 Gy SPE irradiated cardiac tissue. **(b)** Quantification of TGF-β1 in Sham and 0.5 Gy GCR irradiated cardiac tissue. **(c)** Quantification of TGF-β1 in Sham and 1.0 Gy SPE irradiated cardiac tissue. The bar charts are given as median + IQR. GCR: galactic cosmic radiation; HU: hindlimb unloaded; IQR: interquartile range; NL: normal loaded; SPE: solar particle event; TGF-β1: transforming growth factor beta 1; * = p<0.05; ** = p<0.01

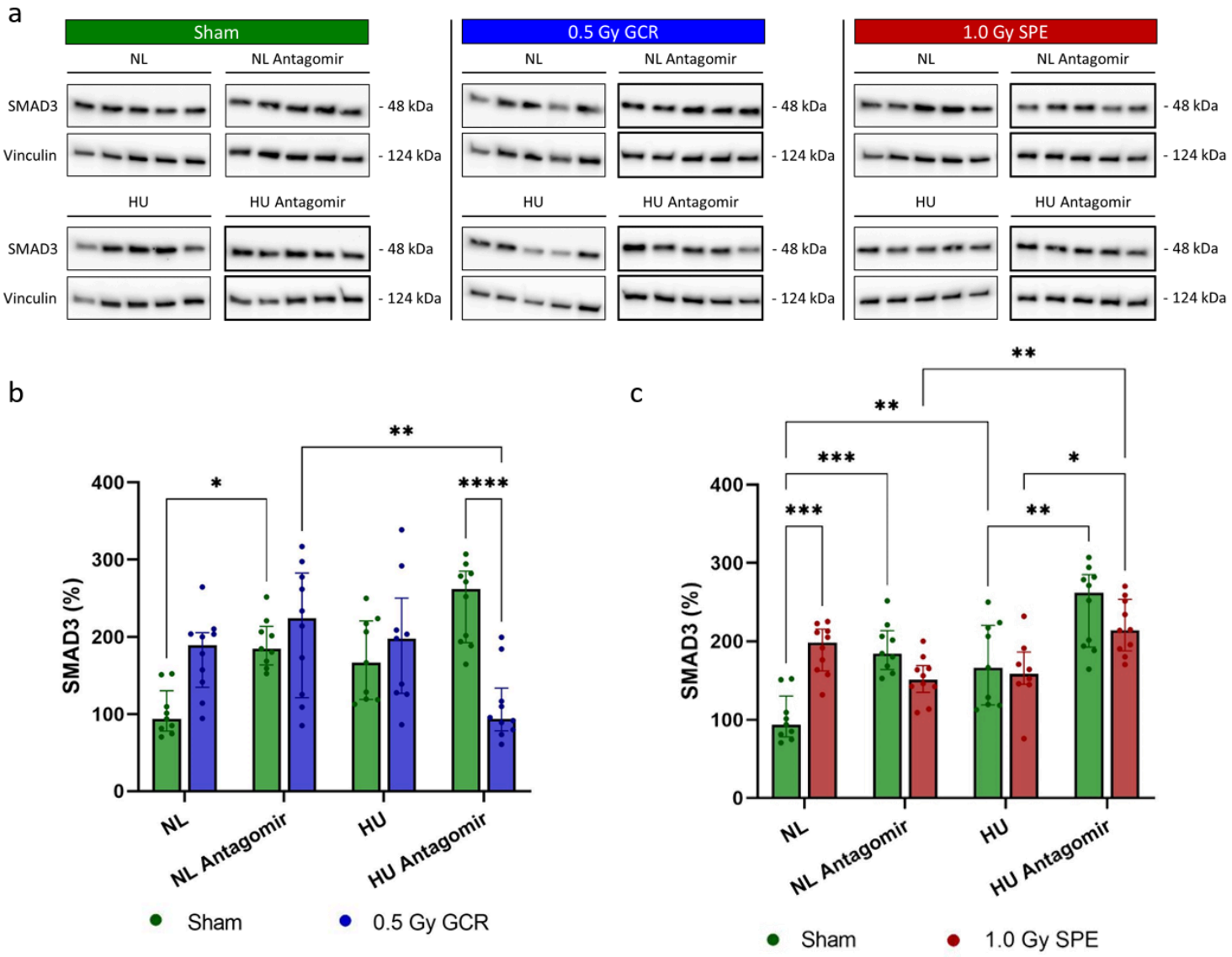


Figure 8

Quantification of SMAD3 by western blot analysis. **(a)** Western blot analysis of SMAD3 in Sham and 0.5 Gy GCR or 1.0 Gy SPE irradiated cardiac tissue. **(b)** Quantification of SMAD3 in Sham and 0.5 Gy GCR irradiated cardiac tissue. **(c)** Quantification of SMAD3 in Sham and 1.0 Gy SPE irradiated cardiac tissue. The bar charts are given as median + IQR. GCR: galactic cosmic radiation; HU: hindlimb unloaded; IQR: interquartile range; NL: normal loaded; SMAD3: mothers against decapentaplegic homolog 3; SPE: solar particle event; * = p<0.05; ** = p<0.01; *** = p<0.001; **** = p<0.0001

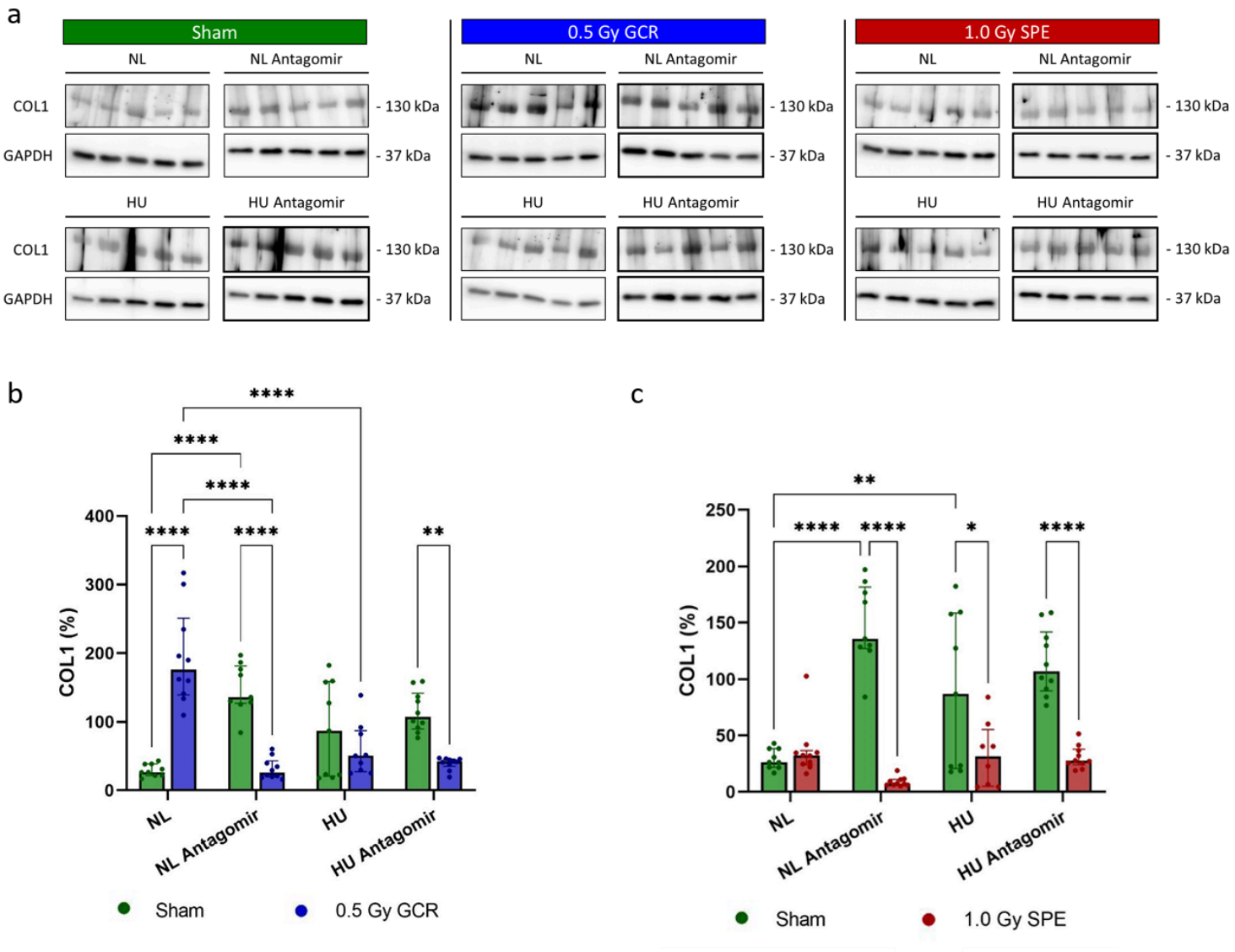


Figure 9

Quantification of COL1 by western blot analysis. (a) Western blot analysis of COL1 in Sham and 0.5 Gy GCR or 1.0 Gy SPE irradiated cardiac tissue. (**b)**Quantification of COL1 in Sham and 0.5 Gy GCR irradiated cardiac tissue. **(c)** Quantification of COL1 in Sham and 1.0 Gy SPE irradiated cardiac tissue. The bar charts are given as median + IQR. COL1: collagen 1; GCR: galactic cosmic radiation; HU: hindlimb unloaded; IQR: interquartile range; NL: normal loaded; SPE: solar particle event; * = p<0.05; ** = p<0.01; *** = p<0.001; **** = p<0.0001

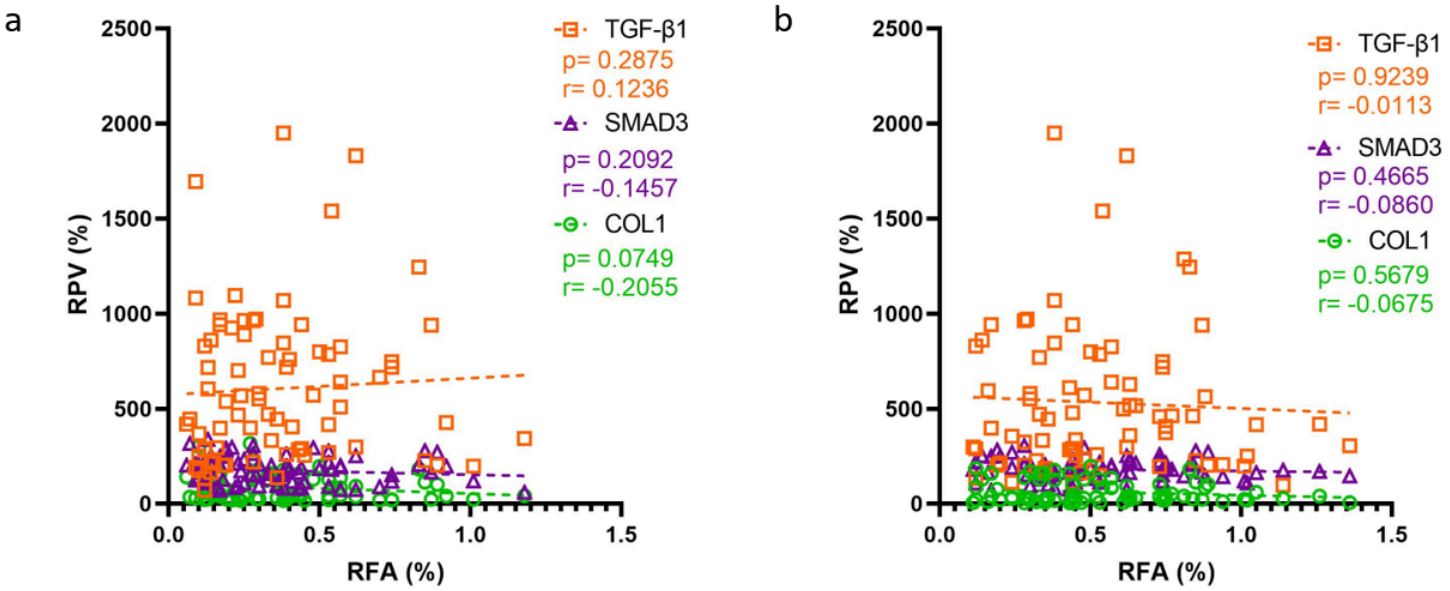


Figure 10

Spearman correlation analysis of RFA and RPV of fibrosis-associated proteins. a Spearman correlation analysis of RFA and fibrosis-associated proteins in 0.5 Gy GCR irradiated mice and Sham. **b** Spearman correlation analysis of RFA and fibrosis-associated proteins in 1.0 Gy SPE irradiated mice and Sham. GCR: galactic cosmic radiation; HU: hindlimb unloaded; NL: normal loaded; SPE: solar particle event

Supplementary Files

This is a list of supplementary files associated with this preprint. Click to download.

- TableS1.xlsx
- TableS2.xlsx